

Figure 4: Superpowers' GUI interface

Latest version: 1.9.4

Official website: www.panda3d.org

Superpowers

Superpowers is a powerful, open source and free development platform enabling developers to create fully customised 2D and 3D games that are highly flexible. It is a cross-platform development tool and supports Windows, Linux and OS X operating systems. It makes use of Typescript to write gaming logic and to highlight the syntax, which simplifies development.

Features

- Easy and well laid GUI interface helps even newbies to quickly learn game development.
- Has a powerful Typescript editor fully packed with features like syntax highlighting, auto completion of code and live error reporting.
- Comes with hundreds of inbuilt licence-free sprites, 3D models, sound effects, fonts and music.
- Built-in library of games and examples acts as a strong platform for beginners.

Latest version: 4.0

Official website: <http://superpowers-html5.com>

MonoGame

MonoGame is powerful free software that Windows based developers and Windows Phone gamers use to run on other systems. It is a cross-platform game development tool and supports Linux, OS X, Android, PlayStation Mobile, Nintendo Switch, etc.

It is basically an open source implementation of the Microsoft XNA 4 framework. The basic objective of MonoGame is to 'write once, play everywhere'.

The following are the technologies that power the MonoGame API's cross-platform capabilities:

- *OpenTK*: a low level C# library that combines OpenGL, OpenCL and OpenAL for 3D graphics.

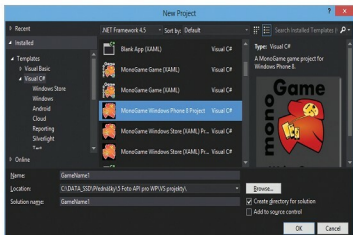


Figure 5: MonoGame user interface

- *SharpDX*: An open source implementation of DirectX API for .NET, which supports high performance of 2D and 3D games and real-time sound.
- *Lidgren.Network*: This is a network library for the .NET framework, which makes use of the UDP socket to provide APIs for connecting to the client and server as well as sending and reading messages.

Features

- Via C# and .NET languages, MonoGame enables developers to write reliable and high-performance game code.
- Open source code enables changes and even porting to new platforms.
- Bundled with more than 1000 games, MonoGame can be used for high-end games development.

Latest version: 3.6

Official website: www.monogame.net

References

- [1] <http://compilgames.net/>
- [2] <https://godotengine.org>
- [3] <http://www.cocos2d-x.org>
- [4] <https://deltaengine.net>
- [5] <https://gamua.com/starling>
- [6] www.panda3d.org
- [7] <http://superpowers-html5.com>
- [8] www.monogame.net

By: Prof. Anand Nayyar

The author is assistant professor in the department of computer applications and IT at KCL Institute of Management and Technology, Jalandhar, Punjab. He loves to work and research on open source technologies, cloud computing, sensor networks, hacking and network security. He can be reached at anand_nayyar@yahoo.co.in. You can watch his YouTube videos at youtube.com/anandanayyar.